

Sets, sets everywhere

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What is algebra about?

Propositions and logics

Conditionals

- $p \rightarrow q$
- True if:
 - p is true, q is true.
 - p is false, q is true.
 - p is false, q is false.
- False if:
 - p is true, q is false.

Vacuous truth

- If Abiria has two girlfriends, then the earth is flat.
- If x is a real number and $x^2 + 4 = 0$, then x is an odd number.
- An empty set is a subset of any set.
 - Let A be a set.
 - $x \in \emptyset \rightarrow x \in A$

Vacuous truth

- $f : \emptyset \rightarrow Y$ is a function.
 - For a function to be well-defined, every element in the domain must have a unique image.
 - i.e., $x \in X \rightarrow x$ has an image in f .
- $f : \emptyset \rightarrow Y$ is an injective function.
 - Injective: $x, y \in X \rightarrow f(x) = f(y) \rightarrow x = y$.

Vacuous truth

- $\emptyset$ is linearly independent subset.
 - ▶ A set is said to be linearly independent, if the only solution to $\sum_i c_i v_i = 0$ is $c_i = 0$ for all i .
 - ▶ $v_i \in \emptyset \rightarrow \sum_i c_i v_i = 0 \rightarrow c_i = 0$

Sets

Set-builder notation

Comprehension form

- $X = \{x \mid x \in U, P(x)\}$
- Or alternatively, $X = \{x \in U \mid P(x)\}$

Comprehension + replacement form

- $X = \{f(x) \mid x \in U, P(x)\}$
- $\forall x (x \in X \leftrightarrow \exists x' (x' \in U \wedge P(x') \wedge x = f(x')))$

Examples

- $A_1 = \{x^2 \mid x \in \mathbb{Z}\}$
- $A_2 = \{x \in \mathbb{R} \mid -2 < x, x < 10\}$
- $A_3 = \{x \in \mathbb{R} \mid x \neq 0\}$
- $A_4 = \{(x, y) \mid x \in \mathbb{R}, y \in \mathbb{R}\}$
- $A_5 = \{2x + 3y \mid x \in \mathbb{Z}, y \in \mathbb{Z}\}$
- $A_6 = \{x \in \mathbb{C} \mid x^4 = 1\}$
- $A_7 = \{(x, \sin(x)) \mid x \in \mathbb{R}\}$

Sets

- $A_8 = \{x \in \mathbb{C} \mid |x| = 1\}$
- $A_9 = \{\{x\} \mid x \in \mathbb{N}\}$
- $A_{10} = \{A \in \mathcal{P}(\mathbb{N}) \mid \forall x \in A (x \text{ is prime})\}$

Subsets

- $A \subset B := \forall x (x \in A \rightarrow x \in B)$

Examples

- $A = \{x \in \mathbb{Z} \mid x \text{ is a multiple of } 10\}$
 $B = \{x \in \mathbb{Z} \mid x \text{ is a multiple of } 100\}$
 - $A \subset B$
- $A = \{x \in \mathbb{C} \mid x^{1212} = 1\}, B = \{x \in \mathbb{C} \mid x^{2424} = 1\}$
 - $A \subset B$

Axiom of Extensionality

Axiom 1 (Axiom of Extensionality)

Two sets A and B are equal if and only if $A \subset B$ and $B \subset A$.



Notations

Definition 1

1. $\mathbb{Z}_n = \{x \in \mathbb{Z} \mid x \text{ is non negative, and } x \text{ is less than } n\}$
2. $n\mathbb{Z} = \{nx \mid x \in \mathbb{Z}\}$
3. $A + B = \{a + b \mid a \in A, b \in B\}$



Extensionality: Example

- Is $2\mathbb{Z} = \mathbb{Z}$ true?
- $2\mathbb{Z} = \{2x \mid x \in \mathbb{Z}\}$
- $2\mathbb{Z}$ is a set of all even numbers.
- Every even number x is in $\mathbb{Z}$. i.e., $2\mathbb{Z} \subset \mathbb{Z}$.
- However, 5 is integer but not in $2\mathbb{Z}$.
- Therefore, $2\mathbb{Z} \neq \mathbb{Z}$.

Extensionality: Example

- Is $\mathbb{Z} + \mathbb{Z} = \mathbb{Z}$ true?
- $\mathbb{Z} + \mathbb{Z} = \{x + y \mid x \in \mathbb{Z}, y \in \mathbb{Z}\}$
- $x + y$ is closed under addition.

Extensionality: Example

- $\mathbb{Z} + \mathbb{Z} \subset \mathbb{Z}$:
 - $n \in \mathbb{Z} + \mathbb{Z}$ means there are integers x, y such that $n = x + y$
 - Hence, $n \in \mathbb{Z}$.
- $\mathbb{Z} \subset \mathbb{Z} + \mathbb{Z}$:
 - If $n \in \mathbb{Z}$, then there are integers $n, 0$ such that $n = n + 0$.
 - Hence, $n \in \mathbb{Z} + \mathbb{Z}$.
- Therefore, $\mathbb{Z} + \mathbb{Z} = \mathbb{Z}$.

Extensionality: Example

- Is $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$ true?
- In other words, is
 - sum of an even number and a multiple of 3 always an integer?
 - any integer is a sum of an even number and a multiple of 3?
- $2\mathbb{Z} + 3\mathbb{Z} \subset \mathbb{Z}$
 - $2n + 3m$ is always an integer.

Extensionality: Example

- Can $2n + 3m$ be 1?
- If so, $2n + 3m = 1$ for some integers n, m .
- If x is an integer, we get $x = x \cdot 1$.
- $x = x(2n + 3m)$
- $x = 2(nx) + 3(mx)$
- Each nx and mx is an integer. Thus, $\mathbb{Z} \subset 2\mathbb{Z} + 3\mathbb{Z}$.
- Therefore, $2\mathbb{Z} + 3\mathbb{Z} = \mathbb{Z}$.